The graphs

Generated from each graph's `meta` and the files a run made. Pictures are real outputs. Regenerate with `claude-debug/make_catalogue.py` (scratch, untracked) after a run.

One craft each

Each graph uses one craft app (plus PlumCraft's building blocks, such as Write Files).

Shelf-pin drilling template

`graphs/crafts/cadcraft/shelf-pin-template.json` · engines: cadcraft · 26 nodes, 25 wires

CadCraft only. A 280 × 900 mm bookcase side with two columns of 5 mm shelf-pin holes on the 32 mm system, 37 mm in from each edge, dimensioned and labelled on separate layers. The DWG is true scale, for a CNC or for tiling a 1:1 print from a CAD program; the PDF and PNG are fitted to the page as a shop drawing (the 100 mm bar shows the scale of any printout). Change the panel size or pitch in the nodes to fit your cabinet; the hole count comes out as JSON as a check.

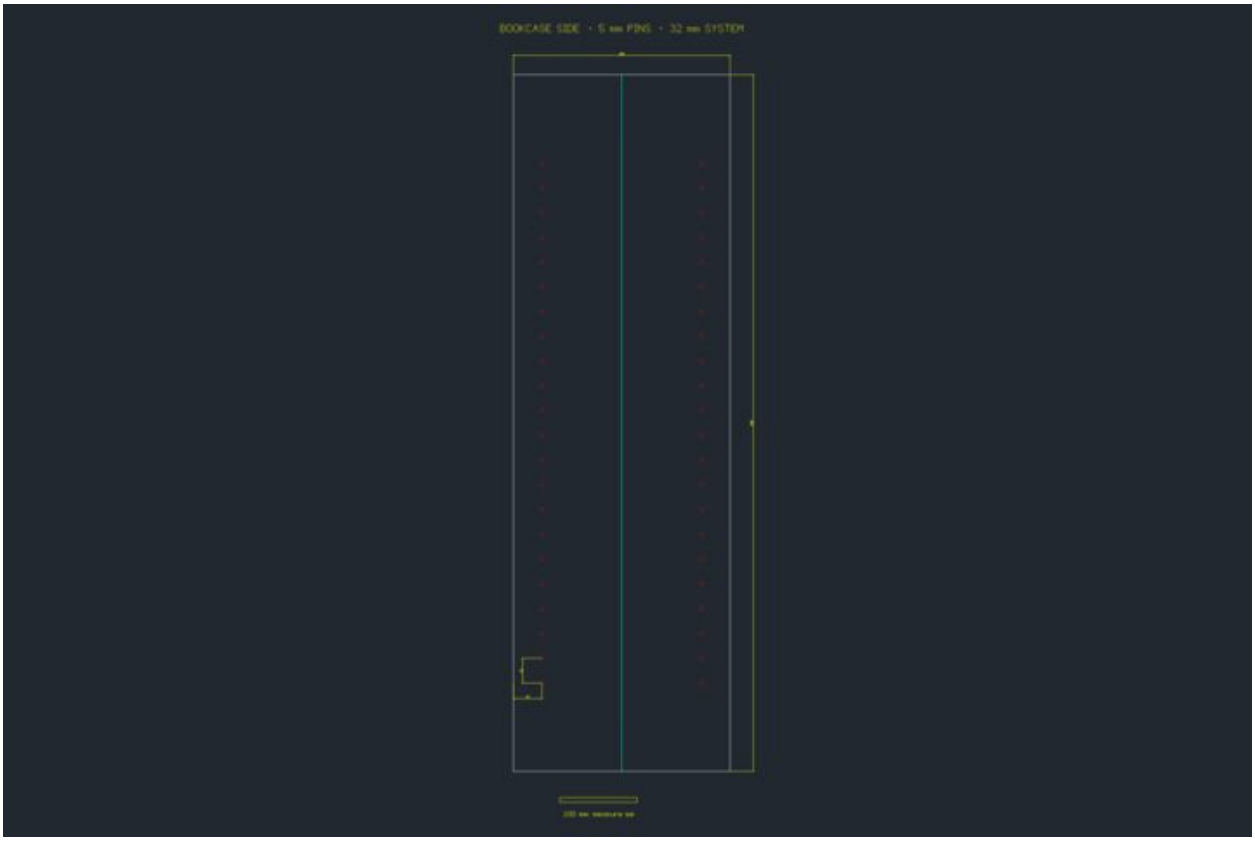


Figure 1: Shelf-pin drilling template

Writes: `hole-count.json`, `shelf-pins.dwg`, `shelf-pins.pdf`, `shelf-pins.png`, `shelf-pins.svg`

Try: Change the circle radius to 3.5 mm for 7 mm pins, or the array rows for a taller panel.

A pitch deck in one outline

`graphs/crafts/deckcraft/cat-cafe-pitch.json` · engines: deckcraft · 17 nodes, 16 wires

DeckCraft only. A Markdown outline becomes a themed, numbered deck with transitions and speaker notes, then a PDF, a PowerPoint file and an animated GIF to post. The startup is a cat café inside a data centre. Edit the outline and every export follows.



Figure 2: A pitch deck in one outline

Writes: `purrfect-uptime.gif`, `purrfect-uptime.pdf`, `purrfect-uptime.png`, `purrfect-uptime.pptx`

Try: Rewrite the outline; theme Nocturne can become Harbor, Ember, Meadow, Paper, Coral Reef or Slate.

Lost: one sock

`graphs/crafts/designcraft/lost-sock-poster.json` · engines: designcraft · 33 nodes, 32 wires

DesignCraft only. A Letter-size lost-property poster with a plum swatch, a big headline, a description, an equation for the statistically inevitable, and eight tear-off phone tabs along the bottom, each rotated a quarter turn. Preflight reports any problems before it goes to the printer, then a PDF/X-4 and a PNG proof come out.

Writes: `lost-sock.pdf`, `lost-sock.png`, `preflight.json`

Try: Replace the text frames; the tear-off tabs are rotated by id, so add frames after them, not before.

A lower third for the evening news

`graphs/crafts/effectcraft/lower-third.json` · engines: effectcraft · 19 nodes, 18 wires

EffectCraft only. A broadcast lower third: a plum bar wipes in from the left, the name fades up over it, and a thin accent line breathes with an expression. Three frames come out (as it starts, mid-move and settled) so you can check the animation without rendering video. Rename the expert in the Text layer nodes.

Writes: `layers.json`, `lower-third-0.15s.png`, `lower-third-0.60s.png`, `lower-third-2.50s.png`

Try: Change the name and role in the two Text layer nodes.

LOST

One sock. Left foot, probably. Grey with a hole where the big toe lives.

Last seen entering the washing machine on Saturday with its partner. The partner came out. It did not.

Answers to nothing. Very attached to me, less so to the other sock.

The science:

$$P(\text{sock}_{\text{lost}}) = 1 - \prod_{i=1}^{n} (1 - p_{\text{wash}})$$

REWARD: the other sock (no longer needed).

Morgan 555-0142

Morgan 555-0142
Morgan 555-0142
Morgan 555-0142
Morgan 555-0142
Morgan 555-0142
Morgan 555-0142
Morgan 555-0142

Figure 3: Lost: one sock

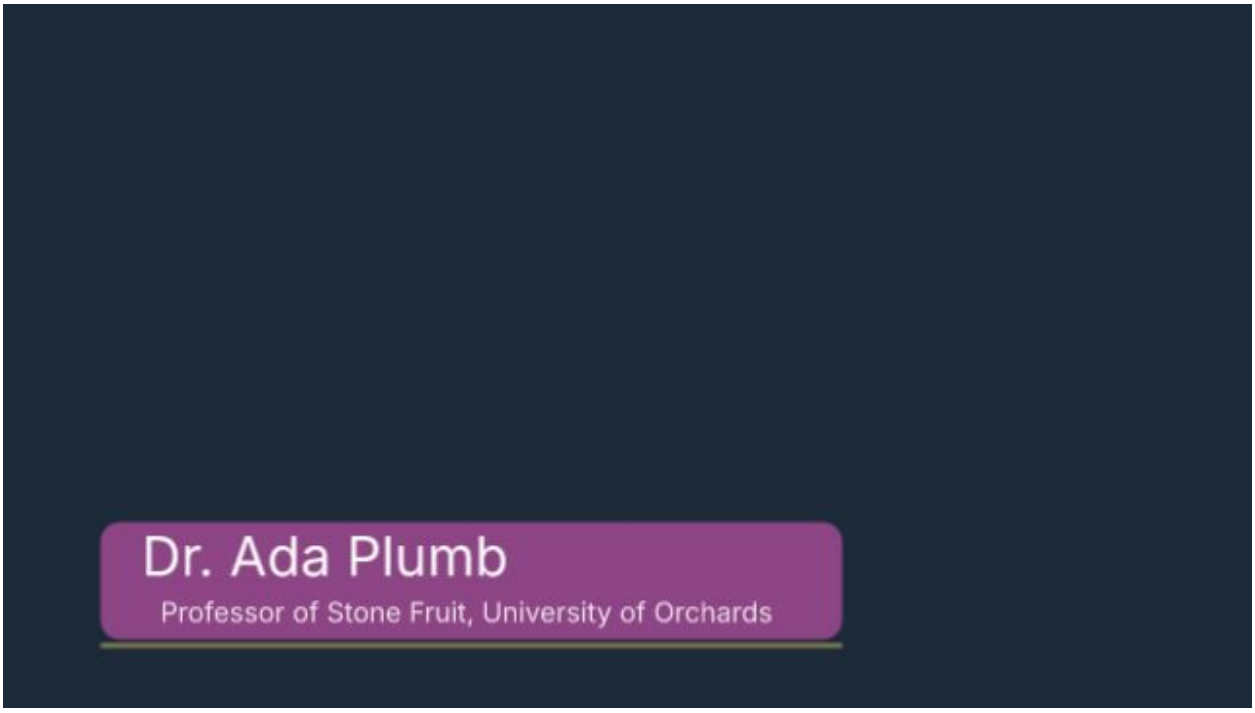


Figure 4: A lower third for the evening news

A design you can paste into code

`graphs/crafts/figcraft/design-to-code.json` · engines: figcraft · 19 nodes, 18 wires

FigCraft only. An onboarding card for a plant-watering app (‘Your fern misses you’), built from frames, shapes and text with auto layout and a soft shadow, exported as PNG and SVG, and as CSS, SwiftUI and React from Inspect code. The JSON files are the code.

Writes: `card-css.json`, `card-react.json`, `card-swiftui.json`, `onboarding-card.png`, `onboarding-card.svg`

Try: Add languages: html, tailwind, vue, compose, leptos, arkui.

A studio ident in ten seconds

`graphs/crafts/filmcraft/studio-ident.json` · engines: filmcraft · 16 nodes, 16 wires

FilmCraft only. Plum Pictures’ opening ident: demo sunset footage on a sequence, a Golden Dusk look, a title over it, then an MP4, a GIF for the group chat and a poster frame. Swap the scene (Aurora, CityNight, Dunes, Plasma, Forest) or the words.

Writes: `plum-pictures-frame.png`, `plum-pictures.gif`, `plum-pictures.mp4`, `sequence.json`

Try: Scenes: OceanSunset, Aurora, CityNight, Dunes, Plasma, Forest. Looks: Night Exterior, Steel Blue, Faded Print…

Is the espresso machine worth it?

`graphs/crafts/gridcraft/espresso-break-even.json` · engines: gridcraft · 21 nodes, 20 wires

GridCraft only. Change the inputs in Write inputs (machine price, cups a week, what the café charges) and the workbook works out how many months until a home espresso machine pays for itself, charts café against home spending over two years, and writes a PDF, a CSV and the break-even month as JSON.

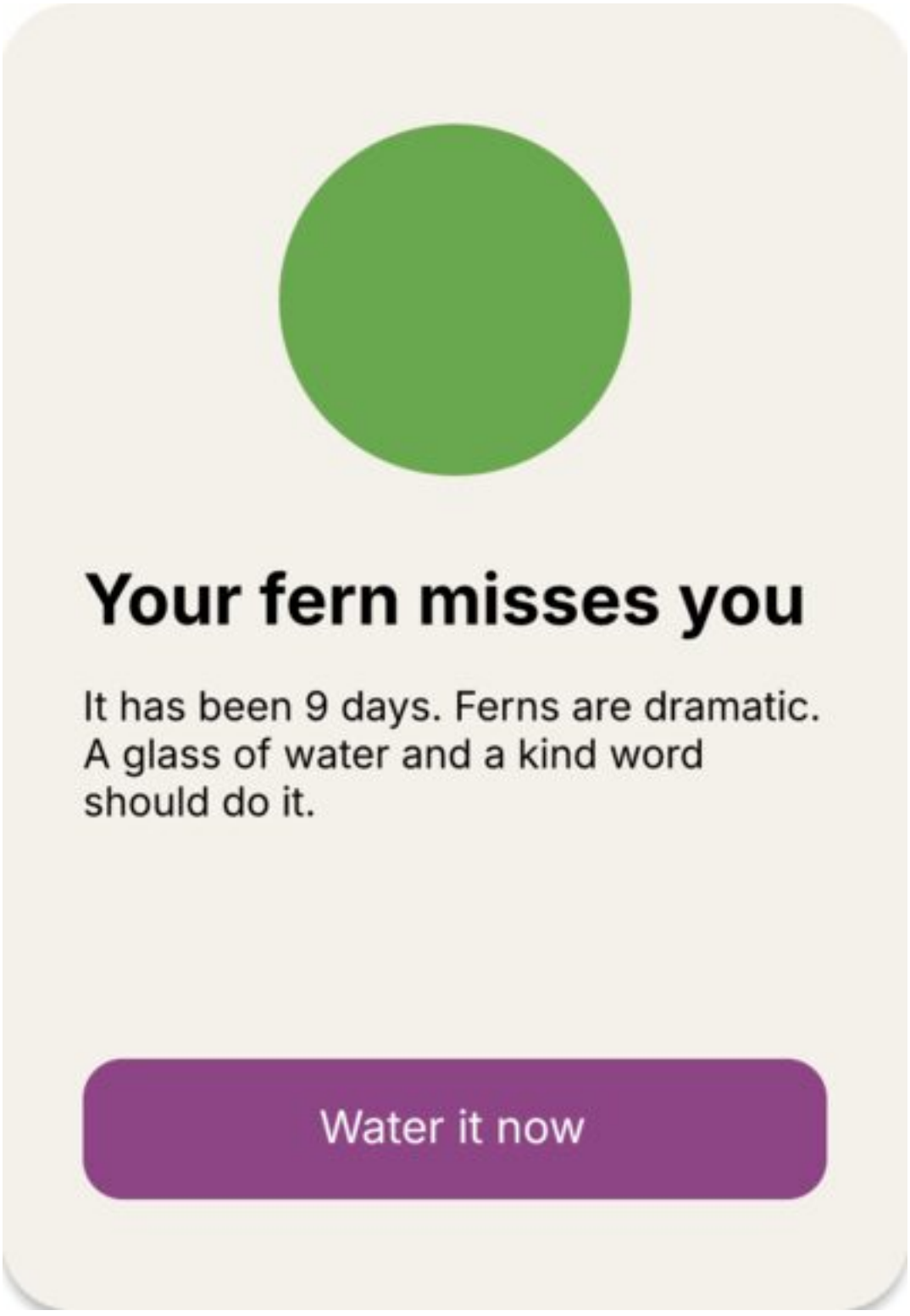


Figure 5: A design you can paste into code



Figure 6: A studio ident in ten seconds

Writes: `break-even-months.json`, `espresso.csv`, `espresso.pdf`

Try: Set cups per week to 2 and watch the break-even month run past the end of the chart.

Read the fine print

`graphs/crafts/pdfcraft/fine-print.json` · engines: pdfcraft · 17 nodes, 16 wires

PdfCraft only. Opens a contract nobody reads (the terms for borrowing a neighbour's lawnmower), counts the words and every 'indemnify', stamps it READ ME, numbers the pages, makes its URLs clickable and checks its accessibility. The JSON answers the question you should ask of any contract: how many times does it make you pay for things?

Writes: `accessibility.json`, `document-info.json`, `indemnify.json`, `terms-annotated.pdf`, `terms.txt`, `word-count.json`

Try: Point Open at any PDF you are about to sign and change Find text to 'liable', 'perpetual' or 'arbitration'.

Plums, five ways

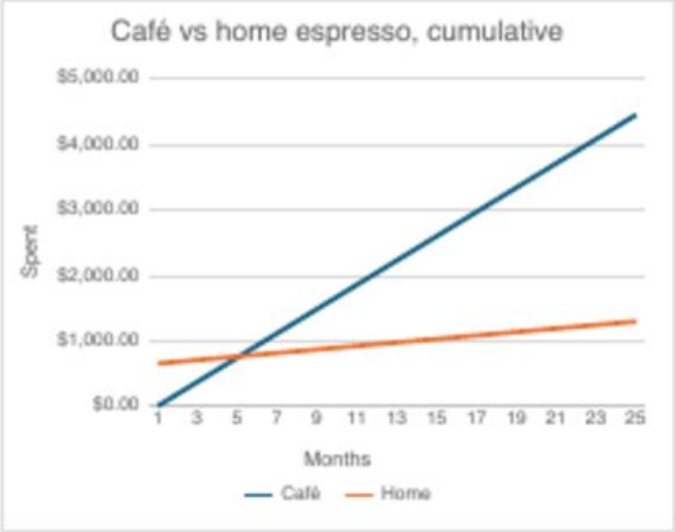
`graphs/crafts/photocraft/plums-five-ways.json` · engines: photocraft · 27 nodes, 26 wires

PhotoCraft only. One photograph of Santa Rosa plums, five looks: an oil painting, gritty black and white, a 1970s cross-processed print, a four-colour pop-art poster, and a moonlit night. Also asks the face detector whether it sees faces in the fruit (pareidolia check). Photo: 'Santa Rosa Plums' by SummerTomato, CC BY 2.0, cropped (assets/CREDITS.md).

Writes: `faces-in-the-fruit.json`, `plums-night.jpg`, `plums-noir.jpg`, `plums-oil.jpg`, `plums-pop.jpg`, `plums-seventies.jpg`

Try: Open your own photo; each look is a short chain you can reorder.

Input	Value
Machine price	$649.00
Beans per cup	$0.45
Milk per cup	$0.15
Café price per cup	$4.75
Cups per week	9
Upkeep per year	$40.00
Break-even month	4.1
Saved after 2 years	$3,155.40



Month	Café	Home
0	$0.00	$649.00
1	$185.25	$675.73
2	$370.50	$702.47
3	$555.75	$729.20
4	$741.00	$755.93
5	$926.25	$782.67
6	$1,111.50	$809.40
7	$1,296.75	$836.13
8	$1,482.00	$862.87
9	$1,667.25	$889.60
10	$1,852.50	$916.33
11	$2,037.75	$943.07
12	$2,223.00	$969.80
13	$2,408.25	$996.53
14	$2,593.50	$1,023.27
15	$2,778.75	$1,050.00
16	$2,964.00	$1,076.73
17	$3,149.25	$1,103.47
18	$3,334.50	$1,130.20
19	$3,519.75	$1,156.93
20	$3,705.00	$1,183.67
21	$3,890.25	$1,210.40
22	$4,075.50	$1,237.13
23	$4,260.75	$1,263.87
24	$4,446.00	$1,290.60

Figure 7: Is the espresso machine worth it?

Terms and Conditions for Borrowing My Lawnmower

Version 3.1, effective immediately and retroactively

1. Definitions

"The Mower" means the red petrol lawnmower kept in my shed, including its grass box, its pull cord, and its personality. "You" means the neighbour who is borrowing The Mower. "Me" means the owner, who is watching from the window.

2. Use of The Mower

You may use The Mower to cut grass. You may not use it to cut hedges, mulch leaves for a "science project", or settle disputes. You agree to indemnify Me against any claim arising from the noise you make before 9 a.m. on a Sunday.

3. Fuel

The Mower is lent with fuel and must be returned with fuel. "Fuel" means petrol, not "the last of the jerry can and some optimism". Questions about fuel may be sent to https://example.com/mower-fuel.

4. Return

The Mower must be returned clean, upright, and within 48 hours. A mower returned late will accrue a fee of one homemade biscuit per day. You agree to indemnify Me for the emotional cost of an empty shed.

5. Damage

If The Mower is damaged, You will tell Me before I find out. You will indemnify and hold harmless Me, my shed, and the gnome by the shed door.

6. Disputes

Disputes will be settled over the fence, in good faith, with tea. These terms are governed by the law of the cul-de-sac. The full archive of previous versions is at https://example.com/mower-terms.

7. Acceptance

By touching The Mower You accept these terms. By reading this far You have done better than most people.

READ ME. NOBODY DOES.

Figure 8: Read the fine print



Figure 9: Plums, five ways

A die you can 3D print

`graphs/crafts/solidcraft/six-sided-die.json` · engines: solidcraft · 37 nodes, 36 wires

SolidCraft only. A 16 mm six-sided die: a cube intersected with a sphere for the classic rounded corners, then 21 pips cut as spheres with opposite faces adding to seven. Exports STL and 3MF for the slicer, STEP for CAD, two renders, and the measured volume so you know how much filament it takes. Change the size or pip radius in the nodes.

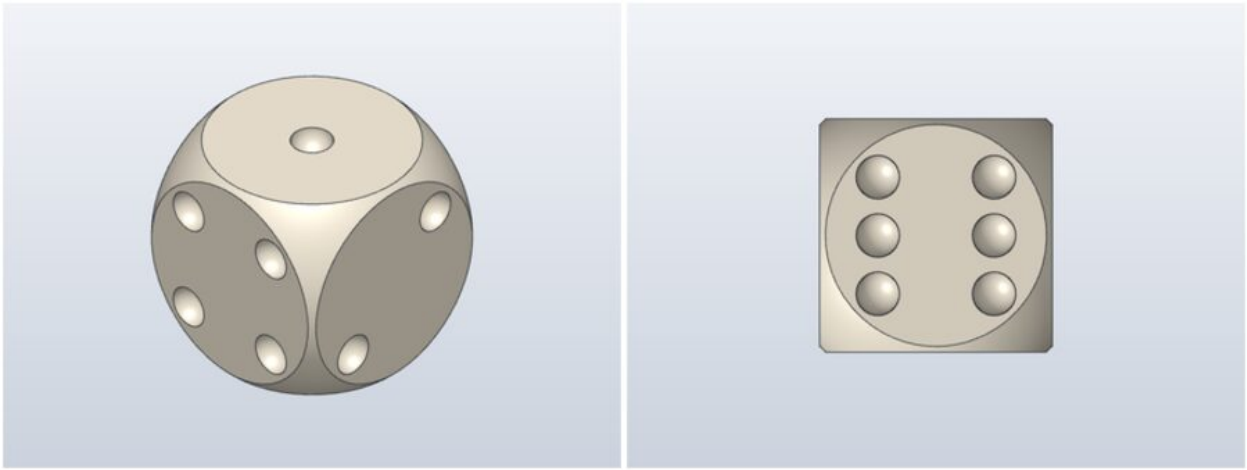


Figure 10: A die you can 3D print

Writes: `die-16mm.3mf`, `die-16mm.step`, `die-16mm.stl`, `die-bottom.png`, `die-iso.png`, `die-measure.json`

Try: Pip radius 1.5 → 1.8 for bolder pips; the measure JSON tells you what it does to the mass.

A ringtone that isn't annoying

`graphs/crafts/soundcraft/ringtone.json` · engines: soundcraft, audio · 24 nodes, 23 wires

SoundCraft, with the audio engine for the bounce as Audio. Four bars at 104 BPM: a bright synth hook, a bass line, and a drum-synth groove, with a little delay and plate reverb on the hook so it sounds expensive. Bounces a normalized WAV for your phone, and exports MIDI and MusicXML so you can take it into a notation app. Change the notes in the MIDI notes nodes and everything follows. WAV to Audio turns the bounce into Audio, ready for any Audio input; here it's measured with Loudness, and the reading saved.

Writes: `ringtone-loudness.json`, `ringtone.mid`, `ringtone.musicxml`, `ringtone.wav`, `session.json`

Try: The notes are plain JSON (pitch, start_ticks, length_ticks, velocity) at 960 ticks a beat.

One badge, five stickers

`graphs/crafts/vectorcraft/sticker-sheet.json` · engines: vectorcraft · 30 nodes, 29 wires

VectorCraft only. A hand-written SVG plum badge becomes a sheet of sticker variants: a print-shop halftone, a wobbly hand-drawn version, a bulging 'warp' version, a 3D extruded coin, and a two-colour risograph recolour. Each is exported as SVG and PNG; the palette of the original comes out as JSON.



Figure 11: One badge, five stickers

Writes: `palette.json`, `sticker-bulge.png`, `sticker-bulge.svg`, `sticker-coin.png`, `sticker-coin.svg`, `sticker-halftone.png`, `sticker-halftone.svg`, `sticker-handdrawn.png`, `sticker-handdrawn.svg`, `sticker-riso.png`, `sticker-riso.svg`

Try: Open your own SVG logo.

A firm but polite letter

`graphs/crafts/wordcraft/toaster-complaint.json` · engines: wordcraft · 20 nodes, 19 wires

WordCraft only. The letter template, filled in by Replace all: a complaint to a smart-toaster company whose firmware update made every slice of toast say 'HELLO'. Change the replacements and you have your own letter, as PDF, Markdown and a page-one PNG.

Writes: `letter.md`, `letter.pdf`, `letter.png`, `word-count.json`

Try: Point the replacements at your own grievance; the template's other text stays put.

Across engines

Each graph crosses two to five engines; values move between them unchanged, by content handle.

An album cover painted by its own music

`graphs/crossings/album-cover.json` · engines: soundcraft, audio, array, photocraft, figcraft · 33 nodes, 34 wires

SoundCraft → audio engine → array engine → PhotoCraft → FigCraft. SoundCraft plays the title track of 'Pits & Pieces' by The Stone Fruits; the audio engine turns the bounce into a spectrogram; the array engine colours it from plum (quiet) to apricot (loud); PhotoCraft paints it in oils; FigCraft sets the band name and title. Change the notes and the cover changes with them.

Writes: `pits-and-pieces-cover.png`, `pits-and-pieces.wav`, `spectrogram-coloured.png`

Try: Change a chord, or the hue span (90° plum→apricot; try 180° for plum→green).

Bouncing-ball storyboard

`graphs/crossings/bounce-storyboard.json` · engines: effectcraft, array · 46 nodes, 56 wires

EffectCraft → array engine. The first exercise of every animation course: a ball that bounces, squashing as it lands and stretching as it leaves, driven entirely by expressions. EffectCraft renders twelve frames across one second; the array engine lays them out as a storyboard sheet with a frame line, which is how you check timing and spacing without playing anything.

Morgan Crumb
4 Rye Lane · Breadford, VT 05001 · morgan@example.com

November 3, 2026

Customer Care

Head of Firmware, ToastLogic Inc.

1 Crust Plaza, Palo Alto, CA

Dear Customer Care,

Since firmware 4.2 arrived on Tuesday, my ToastLogic Pro has branded the word HELLO onto every slice, including the bagels. I did not ask for this, my children now refuse plain toast, and the cat has started to expect a greeting. Please send a downgrade, or a version that can at least spell GOODBYE, by the end of the month.

Thank you for your time.

Sincerely,

Morgan Crumb

Figure 12: A firm but polite letter

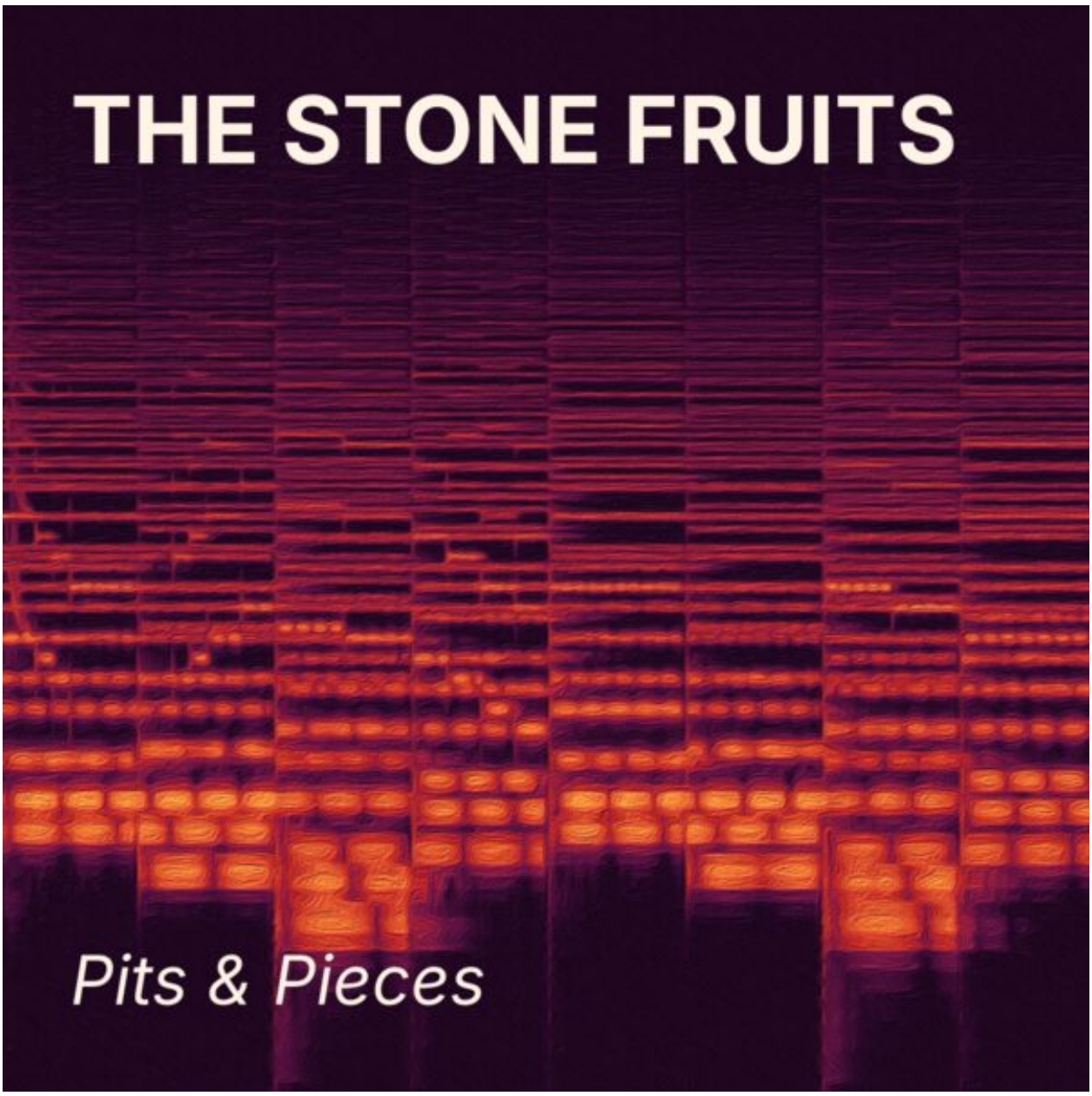


Figure 13: An album cover painted by its own music

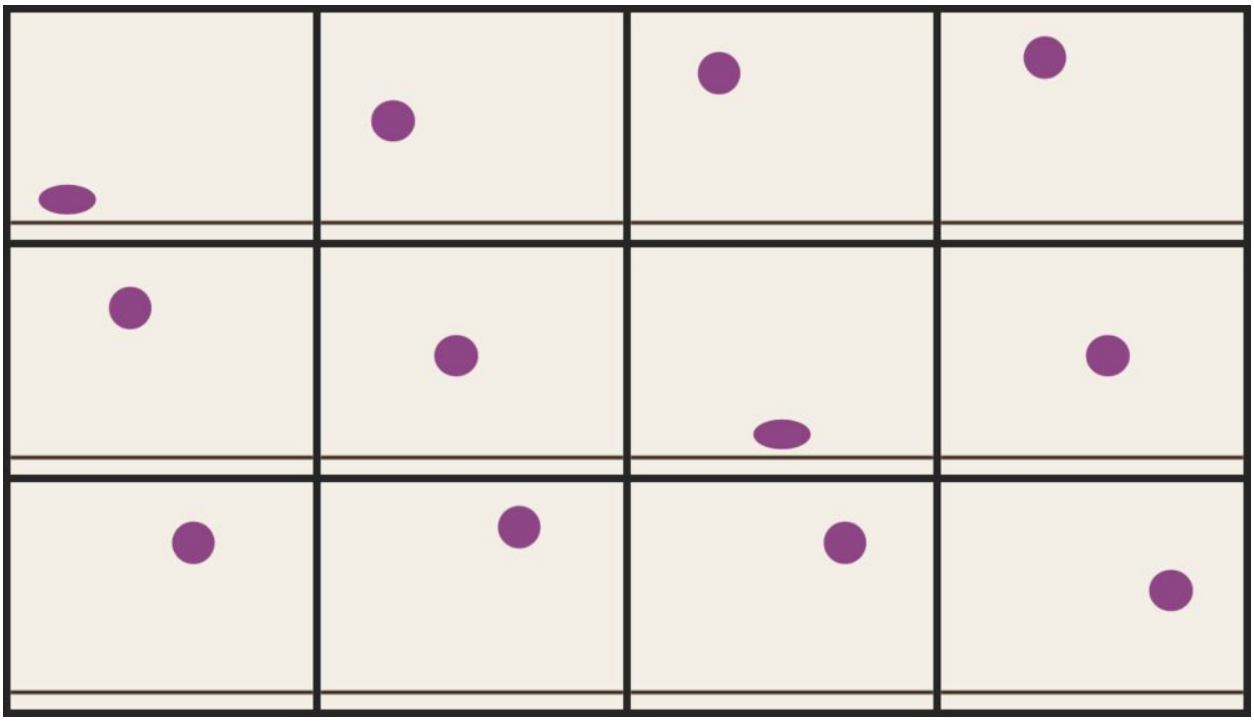


Figure 14: Bouncing-ball storyboard

Writes: `bounce-storyboard.png`

Try: Edit the expressions: the 6 in pow(v, 6) sets how sharp the squash is.

Broken telephone

`graphs/crossings/broken-telephone.json` · engines: array, photocraft, vectorcraft, figcraft · 46 nodes, 52 wires

array → PhotoCraft → VectorCraft → FigCraft → PhotoCraft → VectorCraft → array. The parlour game with pictures: the array engine draws a message, and each craft in turn opens what the last one handed over and passes on its own export: a cheap JPEG, an Image Trace, a half-size PNG, another cheap JPEG, a three-colour trace. The array engine lays every hop side by side and maps what was lost on the way as a heat map. Values cross six engine borders unchanged; the damage is all the crafts' own.

Writes: `every-hop.png`, `what-was-lost.png`

Try: Change the message, or the JPEG qualities; add a hop by copying an Open/Export pair.

Launch day for a die

`graphs/crossings/die-launch.json` · engines: solidcraft, photocraft, figcraft · 42 nodes, 41 wires

SolidCraft → PhotoCraft → FigCraft. SolidCraft models the 16 mm die (cube sphere, 21 pips) in plum and renders it; PhotoCraft gives the render a studio touch-up; FigCraft lays it out as a crowdfunding ad (PNG and PDF). The STL and the measured volume and mass come from the same model, so the ad, the print file and the numbers can't disagree.

Writes: `d6-pro-max-ad.pdf`, `d6-pro-max-ad.png`, `d6-pro-max.stl`, `die-measure.json`

Try: Change the colour in Appearance; ad, STL and render all follow.

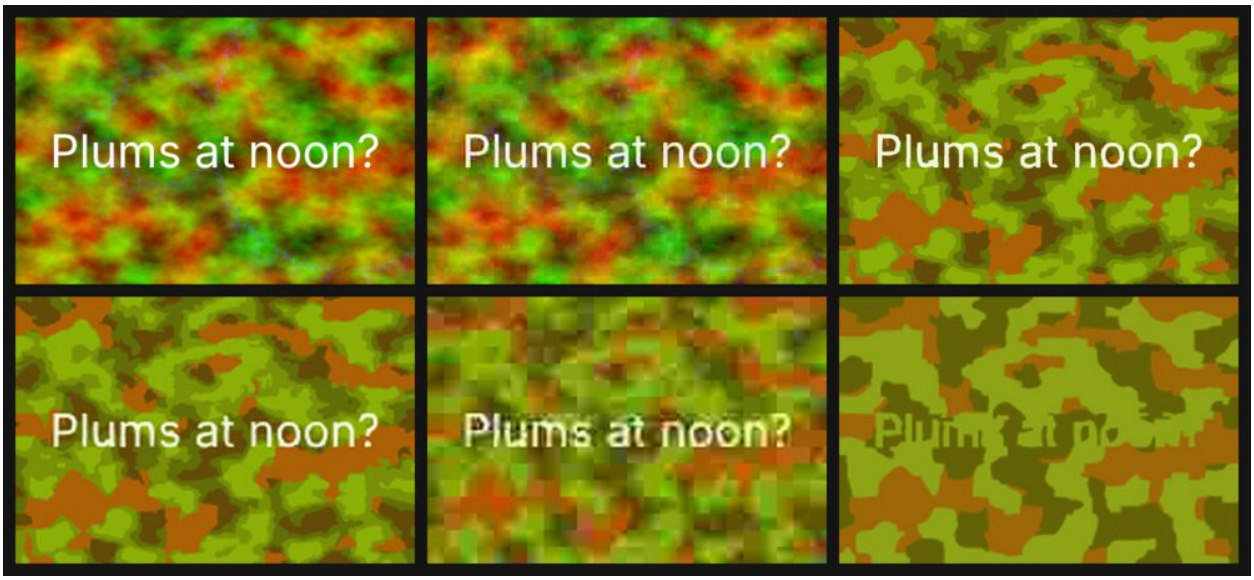


Figure 15: Broken telephone

Is the drummer rushing?

`graphs/crossings/drummer-rushing.json` · engines: soundcraft, audio, gridcraft · 23 nodes, 22 wires

SoundCraft → audio engine → GridCraft. SoundCraft plays sixteen beats at 100 BPM from a drummer who drifts ahead of the click; the audio engine finds every hit in the bounce, and GridCraft lines each one up against where it should have landed, charts the error in milliseconds and gives a verdict. Swap the notes for a real take (Load audio) to judge a real drummer.

Writes: `loudness-lufs.json`, `the-take.wav`, `timing.pdf`, `verdict.txt`

Try: Replace the take: Load audio of a real recording into Transients instead of the SoundCraft bounce.

Garden planner: from a spreadsheet to a plan

`graphs/crossings/garden-planner.json` · engines: gridcraft, cadcraft, vectorcraft · 37 nodes, 39 wires

GridCraft → CadCraft → VectorCraft. The beds live in a spreadsheet (position, size), which also works out the area, the soil each bed needs at 30 cm deep and how many 50-litre bags of compost to buy. GridCraft computes each bed's corners, CadCraft draws them inside the fence as a dimensioned plan, and VectorCraft turns the drawing into a friendly hand-drawn plan to stick on the shed door. Move a bed in the spreadsheet and the plan follows.

Writes: `beds.pdf`, `compost-bags.json`, `plan-dimensioned.pdf`, `plan-for-the-shed.png`, `plan-for-the-shed.svg`

Try: Move a bed in Bed table (x, y, width, depth): the drawing and the compost order follow.

Laser-cut coasters

`graphs/crossings/laser-cut-coasters.json` · engines: cadcraft, vectorcraft, array · 35 nodes, 36 wires

CadCraft → VectorCraft → array engine. CadCraft draws a 95 mm coaster with a ring of twelve vent holes and a hexagon centre; VectorCraft makes it cutter-ready (hairline red cut strokes, outlines offset by half the 0.1 mm kerf); the array engine rasterizes the cut file and labels each separate piece, so you can see at a glance how many parts will fall out of the sheet before you burn any plywood. Outputs the cut SVG and PDF and the parts preview.



Figure 16: Launch day for a die

time	frame	Beat	Should be (s)	Error (ms)
0.597020833	28657	1	0.6	-3
1.194520833	57337	2	1.2	-5.5
1.792020833	86017	3	1.8	-8
2.393270833	114877	4	2.4	-6.7
2.987020833	143377	5	3	-13
3.584520833	172057	6	3.6	-15.5
4.182020833	200737	7	4.2	-18
4.783270833	229597	8	4.8	-16.7
5.377020833	258097	9	5.4	-23
5.974520833	286777	10	6	-25.5
6.572020833	315457	11	6.6	-28
7.173270833	344317	12	7.2	-26.7
7.767020833	372817	13	7.8	-33
8.364520833	401497	14	8.4	-35.5
8.962020833	430177	15	9	-38
9.563270833	459037	16	9.6	-36.7

Mean error (ms)	-20.8
Worst (ms)	-38
Verdict	Rushing: count them in slower

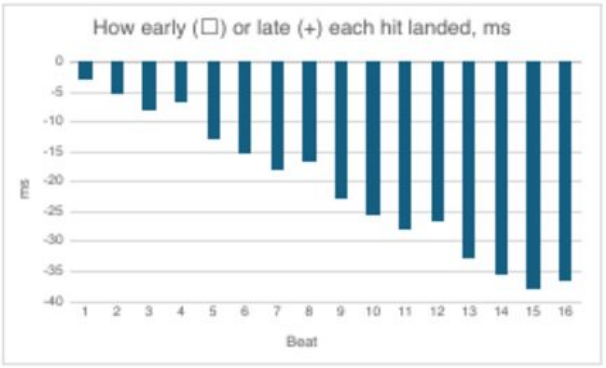


Figure 17: Is the drummer rushing?

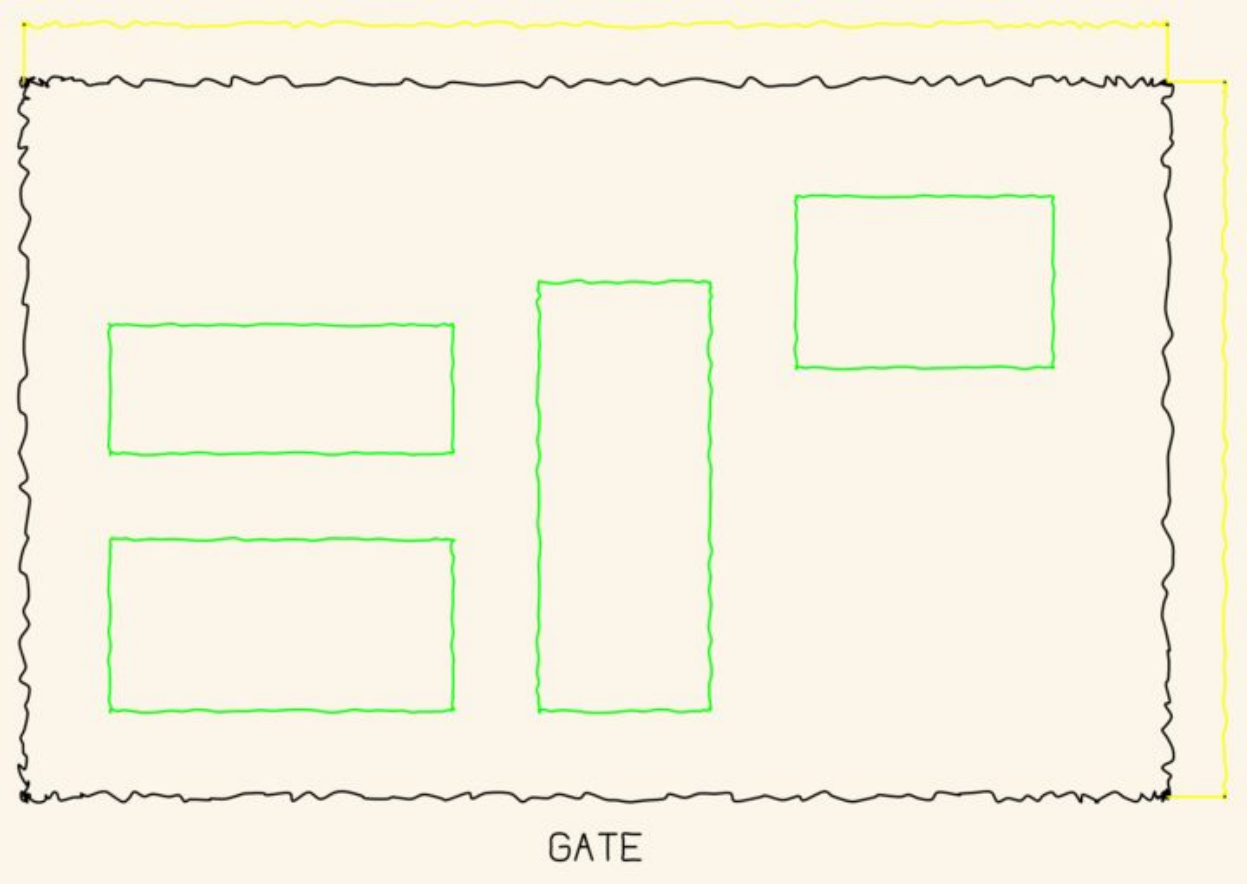


Figure 18: Garden planner: from a spreadsheet to a plan

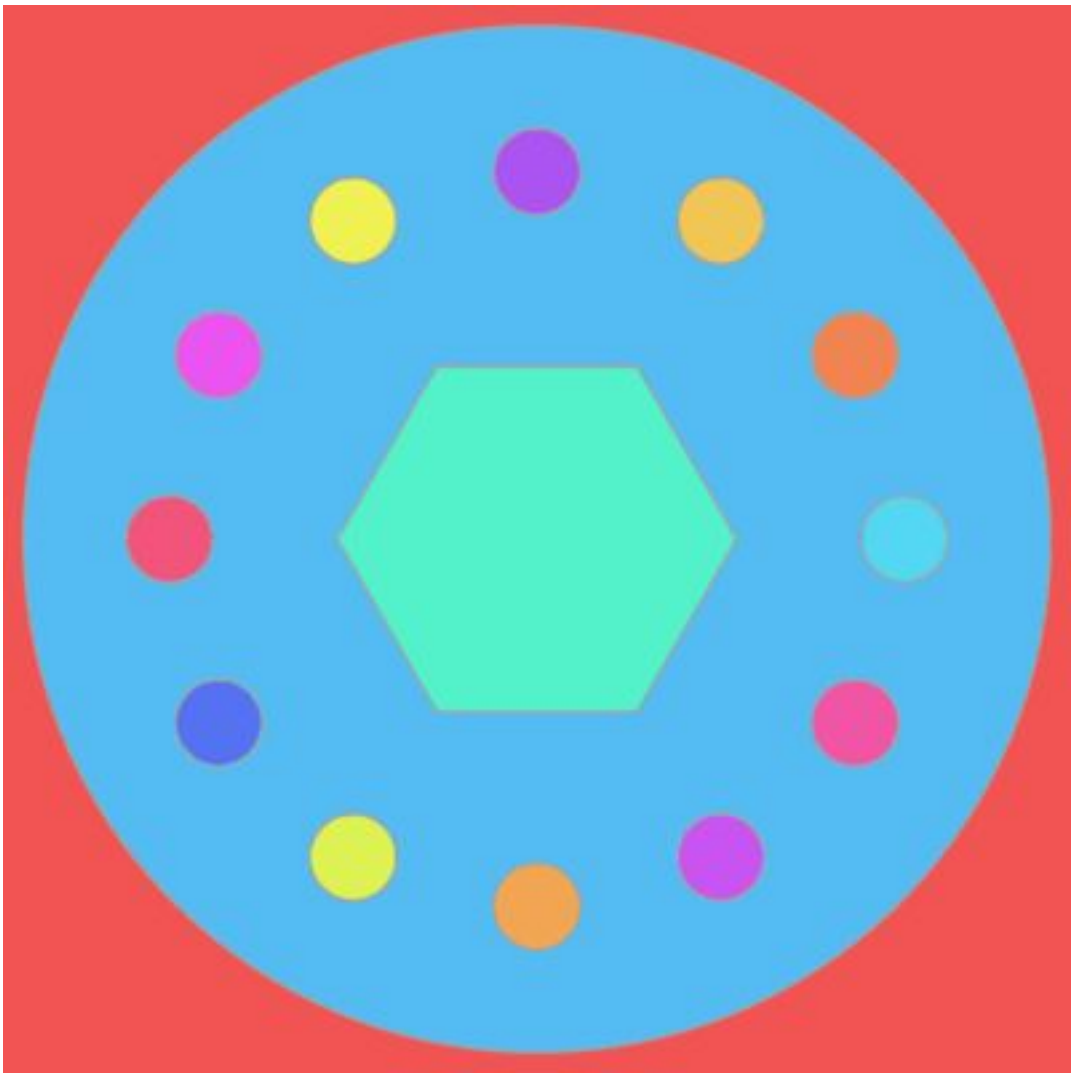


Figure 19: Laser-cut coasters

Writes: `coaster-cut.pdf`, `coaster-cut.svg`, `coaster-drawing.pdf`, `coaster-parts.png`

Try: Change the vent count in Polar array, then count the colours in the parts preview.

Is my photo clipping?

`graphs/crossings/photo-clipping.json` · engines: photocraft, array, gridcraft · 45 nodes, 49 wires

PhotoCraft → array engine → GridCraft. PhotoCraft gives the plums an aggressive 'Instagram' edit (auto tone, +35 brightness, +90 contrast, a hot colour lookup); the array engine takes a 32-bin histogram of each channel of the before and the after; GridCraft charts them and counts how much of each channel is crushed to black or blown to white. The PDF answers 'did I overdo it?' with numbers.

Writes: `histograms.pdf`, `plums-after.png`, `verdict.txt`

Try: Push contrast to 100, or try the noir chain from Plums, five ways, and see the verdict change.

Four sunsets, Warhol-style

`graphs/crossings/warhol-sunsets.json` · engines: filmcraft, vectorcraft, array, photocraft · 35 nodes, 38 wires

FilmCraft → VectorCraft → array engine → PhotoCraft. FilmCraft renders a frame of its ocean-sunset footage; VectorCraft traces it into flat colour and makes four hue-shifted screen prints; the array engine tiles them two by two; PhotoCraft adds paper grain and a faded-film look so it reads as a print. Pick another scene in Demo footage for a different set.

Writes: `four-sunsets.jpg`, `the-frame.png`

Try: Scene Plasma or Aurora; hue shifts in the four Hue/Saturation nodes.

Level	before R	before G	before B	after R	after G	after B
0.015625	3167	3533	2198	14017	13351	10530
0.046875	5074	7093	4036	7714	14266	8778
0.078125	5059	9091	5084	3945	11516	7742
0.109375	5001	11114	5441	2704	12653	6856
0.140625	3586	11638	6454	2680	10692	7029
0.171875	2992	12313	6494	2974	8173	6954
0.203125	2577	11686	7373	2758	8425	6970
0.234375	2976	11843	8351	2883	8168	6319
0.265625	3718	11994	9090	2909	7246	5650
0.296875	4335	12375	9130	2853	6119	5640
0.328125	5024	12778	9383	2716	6214	5580
0.359375	5412	13006	10038	3115	6777	5517
0.390625	6850	13900	10551	2979	5997	5431
0.421875	9327	14597	11234	3444	6308	5172
0.453125	11311	15120	11554	4201	6242	5674
0.484375	12269	14659	11598	4691	4437	6096
0.515625	13582	13449	11786	5401	6631	6896
0.546875	14384	12008	12263	6310	7326	7139
0.578125	14340	10839	13316	7229	7732	7676
0.609375	14015	9710	14234	8103	8684	7571
0.640625	14008	8887	13944	8905	8815	9904
0.671875	13707	7536	13334	9925	8592	12537
0.703125	12871	6729	11883	10340	10406	16563
0.734375	12520	5747	9561	10873	10356	19914
0.765625	11767	4850	8875	11302	10065	26092
0.796875	11451	3302	7817	12323	12114	29985
0.828125	10738	1796	6721	13035	12744	22910
0.859375	9973	795	6429	13407	14650	0
0.890625	8705	246	5745	15195	16209	0
0.921875	7937	111	4366	16650	2217	0
0.953125	6591	239	2716	20436	0	0
0.984375	7858	141	2126	37108	0	0

Figure 20: Is my photo clipping?



Figure 21: Four sunsets, Warhol-style